

Vyapar AI Employee

Independent Architecture Audit — Executive Summary

Review type: Production readiness assessment (Principal Architect / LLM / Memory / Conversation AI)

Date: June 3, 2026

Maturity rating: Level 2.5 Functional MVP → Early Production

Diagnosis verdict: Mostly Correct — missing conversation brain; hypothesis incomplete

Bottom line: The system logs leads well and sometimes remembers names, but it is not a coordinated AI employee. Current architecture is: *regex extractors* → *SQL persistence* → *prompt concatenation* → *hope the LLM behaves*. That demos; it does not reliably sell or converse in Nepali Telegram at production quality.

1. System Under Review

Telegram-based AI employee for Nepali ISP/business use. Components: Intent Engine, Lead Engine, Memory, Knowledge Base, Product Search, Company Profile, Prompt Builder, OpenAI, CRM/leads, Render deployment.

Production logs confirm working: LEAD_SIGNALS_DETECTED, LEAD_UPSERTED, EMPLOYEE_REPLY_GENERATED (and SALES_MODE_ACTIVE after Phase 2 fixes when deployed).

User-visible failures: inconsistent memory, persona drift (sales / FAQ / Wikipedia / memory bot), weak sales conversion, pitch repetition, conversation repair failures, name corruption.

2. Task 1 — Evaluate Your Diagnosis

Verdict: Mostly Correct

Your claim	Assessment
Not Render/GitHub/Cursor/OpenAI model quality	Mostly true — orchestration failure
Missing conversation brain	Correct — <code>ai_employee_engine.py</code> is linear, not a controller
Missing State / Persona / Memory Validation / Repair / Sales Controller / Goal Tracker	Correct — not first-class modules
Lead engine working = architecture OK	Incorrect — CRM logging ≠ sales behavior

Under-weighted in your hypothesis: single-turn LLM; unvalidated regex memory; prompt overload; no response validator; partial deploy drift.

3. Task 2 — Real Root Causes

#	Root cause	Evidence
RC-1	No conversation orchestrator	One function runs entire pipeline; no goal/phase object
RC-2	Single-turn LLM + prompt stuffing	<code>openai_engine.py</code> : only current message as input
RC-3	Memory write eagerly, trust blindly	<code>extract_name</code> matches <code>i am x</code> → "Indian I Am Nepali" bug
RC-4	Persona = prompt text only	7+ prompt blocks compete every turn
RC-5	Sales mode = logging flag	Bool + instructions; no enforced playbook
RC-6	Lead CRM decoupled from dialogue	Score/stage stored; no next-action planner
RC-7	No conversation repair layer	"Puchna haina, sodhna bhannus" not handled

RC-8	Generic knowledge base	Platform docs, not tenant HONS FAQs
RC-9	Release discipline gaps	Partial deploys; no CI golden-path tests

Production evidence (June 3 thread)

User message	Failure	Cause
Mero naam khemraj adhikari ho	Works	Clean declarative regex
Which internet?	Full catalog dump	Non-sales path, include_full_catalog=True
Internet bhaneko ke ho?	Wikipedia answer	No persona scope controller
Puchna haina sodhna... i am indian i am Nepali	Name = "Indian I Am Nepali"; still says puchna	No memory validator; no repair layer

4. Task 3 — Top 10 Architecture Problems

Rank	Problem	Severity	Impact
1	No conversation orchestrator / state machine	Critical	Sales, UX, Conversion
2	Single-turn LLM + prompt stuffing	Critical	UX, Sales, Memory
3	Unvalidated regex memory writes	Critical	Memory, UX
4	No response validator	Critical	Sales, Conversion
5	Persona only in prompts	High	UX, Sales
6	Lead CRM decoupled from planner	High	Conversion
7	No conversation repair layer	High	UX, Memory
8	Prompt block competition	High	Sales, UX
9	Weak tenant knowledge	Medium	UX
10	Deploy/test discipline	Medium	Reliability

5. Task 4 — Failure Contribution Estimates

Category	Est. %
Architecture (no orchestrator)	28%
Conversation logic (single-turn, no planner)	22%
Prompt design (overload, conflicts)	18%
Memory design (regex, no validation)	15%
State management	10%
Lead engine	4%
Persona system	2%
Knowledge base	1%

Key insight: Lead engine is an event logger, not the main failure driver.

6. Task 5 — Why Bugs Reappear

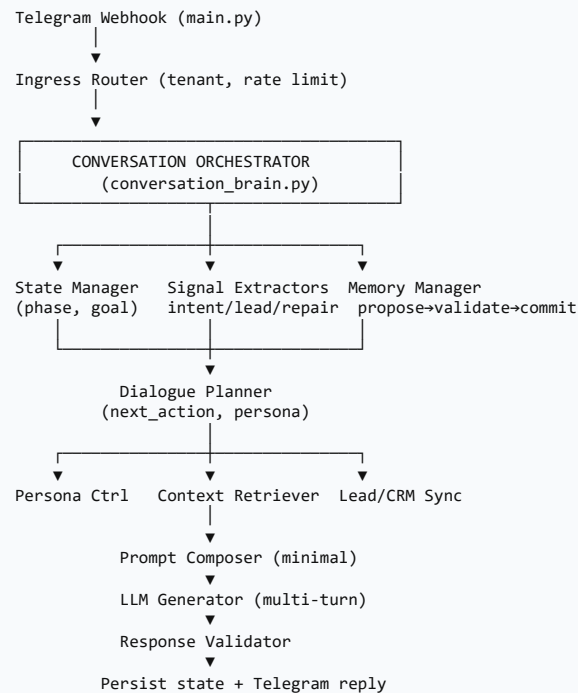
- Fixes patch symptoms in isolated files (regex, prompt hints) without central state
- Each fix adds prompt surface area — instructions compete and dilute
- `sales_mode=True` in logs ≠ correct UX behavior
- No Nepali golden-path regression tests
- Partial deploys: production ≠ repo behavior
- LLM asked to orchestrate what code should own

7. Task 6 — Missing Layers

Layer	Status
State Manager	MISSING
Persona Manager	MISSING
Memory Validator	MISSING
Conversation Repair Manager	MISSING
Sales Controller	PARTIAL (bool + regex only)
Goal Tracker	MISSING
Dialogue Planner	MISSING

8. Task 7 & 8 — Phase 3 Architecture (Recommended)

Principle: Code owns state and decisions; LLM owns language generation.



Sales playbook states: DISCOVER → QUALIFY → PROPOSE → HANDLE_OBJECTION → COLLECT_CONTACT → HANDOFF → CLOSED

9. Task 9 — Implementation Priority

Priority	Work	Why first
P1	Conversation State + Orchestrator	Stops prompt whac-a-mole
P2	Memory Validation Layer	Stops name corruption; high trust impact
P3	Multi-turn LLM + Minimal Prompt Composer	Biggest quality lift per token
P4	Dialogue Planner + Sales Playbook	Turns lead score into actions
P5	Response Validator	Blocks repeat pitch, wrong escalation
P6	Conversation Repair Manager	Handles "sodhna not puchna" corrections
P7	Tenant Knowledge Base	HONS FAQs, policies, talk-tracks
P8	Observability + CI golden tests	20 Nepali scenario tests

10. Task 10 — Maturity Rating

Level	Status
Level 1 Prototype	Passed
Level 2 Functional MVP	Current core
Level 3 Early Production	Partial — deployed, inconsistent under real Nepali dialogs
Level 4 Stable Production	Not met
Level 5 Enterprise Ready	Not met

11. What Works (Keep)

- Telegram webhook + FastAPI shell
- Multi-tenant company_manager + company_profiles.json
- Durable memory, chat history, lead persistence
- Lead extraction/scoring/export
- Phase 2 sales mode + objection patches (on disk)

12. What Must Be Redesigned

- Conversation orchestration (the "brain")
- Memory write path (validation before commit)
- LLM invocation (multi-turn, minimal prompts)
- Output validation before Telegram send
- Persona/phase routing (not prompt-only)

CTO one-liner: The bot can store a lead and store a name, but it cannot reason about what kind of turn this is — correction vs identity vs product vs education — so it corrupts memory and repeats templates. Do not add more regex or prompt text until Conversation State + Memory Validator + Response Validator exist.